INTERNATIONAL MONETARY FUND

SENEGAL

December 27, 2018

STAFF REPORT FOR THE 2018 ARTICLE IV CONSULTATION AND SEVENTH REVIEW UNDER THE POLICY SUPPORT INSTRUMENT AND REQUEST FOR MODIFICATION OF ASSESSMENT CRITERIA—DEBT SUSTAINABILITY ANALYSIS[1]

Approved By
Dominique Desruelle and Maria Gonzalez (IMF), and Paloma Anos-Casero (IDA)

Prepared by the staffs of the International Monetary Fund and the International Development Association

Senegal Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*Low*
Overall risk of debt distress	*Low*
Granularity in the risk rating	*N/A*
Application of judgment	Yes, incorporating ongoing debt operation and availability of guarantees

Senegal has expanded its debt perimeter to include para-public entities and state-owned enterprises (SOEs) and remains at low risk of debt distress despite short-term breaches of two external debt indicators under the most extreme scenarios. The low risk of debt distress is predicated on: (i) ongoing debt liability management, guarantees to address currency risk, access to liquid financial assets and a sound track record of market access; and (ii) adherence to the planned fiscal consolidation path, an acceleration of reforms, and a prudent borrowing strategy. Looking ahead, it will be important to contain fiscal pressures from Treasury operations and address fiscal risks from the broader public sector, including the energy sector.

[1]This DSA has been prepared following the revised LIC-DSA framework. It updates the previous Joint DSA (IMF Country Report No. 18/8). Senegal's debt carrying capacity, calculated based on the April 2018 WEO and the October 2018 WEO and the 2017 CPIA is classified as strong. The applicable thresholds to public and publicly guaranteed external debt are: 55 percent for the PV of debt- to-GDP ratio, 240 percent for the PV of debt-to-exports ratio, 21 percent for the debt service-to-exports ratio, and 23 percent for the debt service-to-revenue ratio. The applicable benchmark for the PV of total public for strong debt carrying capacity is 70 percent of GDP.

BACKGROUND

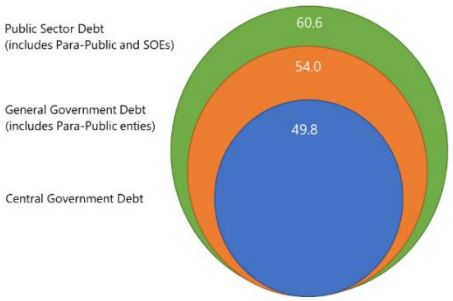
1. In this DSA, the debt perimeter has been expanded to include para-public entities and state-owned enterprises (SOEs) (Box 1). Because of the expanded coverage starting in 2017, the end-2017 debt stock has increased by 10.8 percentage points of GDP to 60.6 percent of GDP, compared to central government debt of 49.8 percent of GDP. The structural break in 2017, combined with the GDP rebasing which resulted in an increase in the level of GDP of about 30 percent, makes comparison with the last published DSA (Country Report No. 18/8)—based only on central government data and the "older GDP base" DSA—somewhat difficult.

Box 1. Coverage of Public Sector Data in the DSA

The DSA presented in this document is based on a broader coverage of the public sector, both for the debt stock and government guarantees, and the fiscal deficit. The public sector includes the (i) central government, (ii) para-public entities which are part of the general government, and (iii) SOEs.

Total public debt data, including debt of para-public entities and SOEs, are provided by the authorities. Revenue and expenditure data on para-public entities and SOEs are derived from technical assistance (TA) reports from the Statistics and Fiscal Affairs Departments of the IMF. Combining the realized deficit of 3.3 percent of GDP for the central government in 2016 with the rest of the public-sector data derived from the TA reports and the authorities, puts the end-2016 overall public-sector deficit at 1.4 percent of GDP. Over the medium term, the implementation of an ambitious public-sector investment program under the *Plan Senegal Emergent* (PSE), including from the national electricity company SENELEC (the largest SOE), PETROSEN (the state-owned hydrocarbon company), and the startup costs from the newly created Air Senegal state-owned airline company, widen the overall public-sector deficit to 2.8 percent of GDP. The expanded coverage of debt data increases the end-2017 debt stock by 10.8 percentage points of GDP pushing the total debt stock to 60.6 percent of GDP compared to central government debt of 49.8 percent of GDP.

Senegal: Total Domestic Public Debt, year end 2017
(Percent GDP)



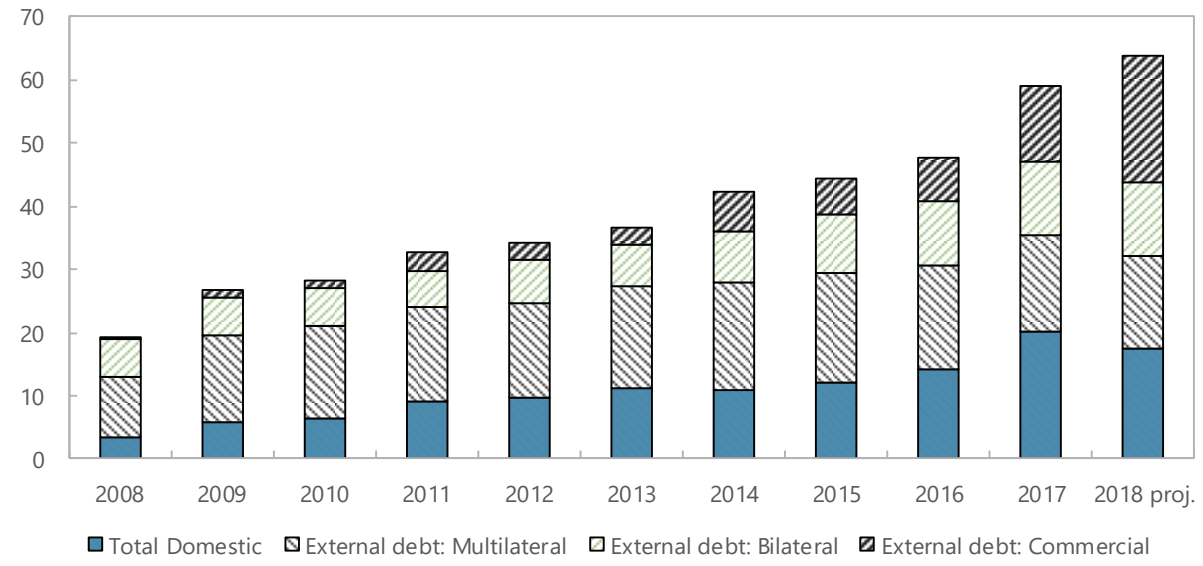
The authorities remain committed to tackling public debt data gaps and improve debt transparency. The expanded coverage of public sector data in this DSA starting in 2017 creates a structural break with historical data, making comparison to the previous DSA more difficult.

2. External debt vulnerability indicators have somewhat deteriorated over the past year, reflecting an increase in external non-concessional borrowing and the coverage of para-public entities and SOEs' debt and guarantees.[2] For 2018, external public debt in Senegal is projected at 47.6 percent of GDP, compared to 45.2 percent projected in the previous DSA. The expansion of the debt

[2] In this DSA, external debt is defined based on currency rather than residency. External debt includes borrowing from the IMF.

perimeter represents one of the key factors in this increase. Also, in 2018, Senegal issued a $2.2 billion Eurobond, its largest ever and twice the previous issuance of $1.1 billion in 2017. Both the 2017 and 2018 issuance substituted, to a large extent, for borrowing on the regional market. Senegal's Eurobond issuance had the benefit of creating space in the regional bond market for WAEMU countries without access to international markets, and, combined with substantial Eurobond issuance by Cote d'Ivoire, helped in the short run to further build up WAEMU pooled reserves, which had fallen in 2016. Nevertheless, the increase in the debt to GDP ratio was held back by the GDP rebasing.

Figure 1. Senegal: Public Sector Debt
(Percent GDP)



Source: Staff estimates

3. The new Eurobond shifted borrowing from the domestic to the external market. Domestic debt is projected to decrease from 20.2 percent of GDP at end-2017 to 16.9 percent of GDP at end-2018. Total public debt, including the expanded perimeter coverage (Box 1), is projected to reach 64.5 percent of GDP in 2018, although this includes accumulation of 2018 budget overfinancing (1.6 percent of GDP) in an escrow account which will be unwound to meet 2019 financing needs. Despite the expanded coverage, the increase in the 2018 debt-to-GDP ratio remains somewhat moderate compared to the previous DSA estimation because of GDP rebasing, as well as a decline in domestic debt given that Senegal has not accessed the regional market in 2018. The higher debt also reflects higher financing requirement owing to the need for the Treasury to finance deficits of the Post Office and the Civil Service Pension, as well as the tapping of unutilized appropriations of past budgets through the *comptes de dépôt*. The high public debt service, which is projected to reach 40.9 percent of revenue in 2018, reflects an increased recourse to non-concessional borrowing in the last two years and the partial repurchase of the 2011 Eurobond.[3]

[3] The repurchase on the market of 40 percent of the bullet repayment due in 2021 was made using the proceeds of the 2018 Eurobond issuance. The difference between the market price and the at par price,is accounted for as increased debt service in 2018.

Table 1.: Evolution Debt Indicators DSA
(Percent GDP)

	2018	
	December 2017 DSA	Current DSA 2018
Public and Publicly Guaranteed Debt		
(Previous GDP)	58.7	
(Rebased GDP)	45.7	64.5
Increase in Debt		18.8
Expansion to include Public Sector Debt		11.7
2018 Eurobond Issuance		7.3
Reduction in domestic debt from external		-0.2

Source: Staff estimates.

Table 2. Senegal: Coverage of Public Sector Debt and Design of the Contingent Liability Stress Test

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	X
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	X
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	X

1	**The country's coverage of public debt**	The central, state, and local governments plus extra budgetary funds, central bank, government-guaranteed debt, non-guaranteed SOE debt		
		Default	**Used for the analysis**	**Reasons for deviations from the default settings**
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	0.0	
4	PPP	35 percent of PPP stoc	2.1	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)		7.1	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

UNDERLYING ASSUMPTIONS AND BORROWING PLAN

4. The authorities remain committed to a debt management strategy that will rely predominantly on concessional and semi-concessional borrowing. They aim to borrow largely from traditional bilateral and multilateral lenders, and resort to non-concessional borrowing only in exceptional cases and for specific high-return projects. Eurobond issuances and, more generally, borrowing on commercial terms, will be considered if financing terms are favorable and if it is not possible to obtain concessional financing from development partners, particularly the African Development Bank and the World Bank.

5. The DSA is consistent with the macroeconomic framework outlined in the Staff Report and updates the previous DSA produced in Country Report No. 18/8, for the Fifth Review of the Policy Support Instrument (PSI). In line with the previous DSA, the baseline scenario assumes the implementation of sound macroeconomic policies, structural reforms, and an ambitious investment plan, as outlined in the *Plan Senegal Emergent* (PSE). It is also assumed that Senegal's fiscal deficit remains at 3 percent of GDP over the long term, consistent with WAEMU targets. This scenario is expected to deliver strong and sustained economic growth and a narrowing fiscal deficit over the medium term. In the current framework, growth, imports, investment, exports and other macroeconomic variables include the implementation of two significantly large off-shore oil and gas projects (the so-called GTA and SNE projects), including pre-production investment and borrowing, with first production and export of oil and gas expected in 2022. As a result, medium-term projections for variables like real GDP growth are more optimistic than historical averages, but aligned with recent outcomes in 2016-17, implying that the take-off is assumed to persist, conditional on reforms being implemented. The main assumptions are as follows:

- **Real GDP growth** is projected to be 6.2 percent in 2018. The growth momentum is maintained in the medium term capturing the effects of infrastructure investment, export growth, and reforms under the PSE. The growth rate peaks at 11.6 percent in 2022, with a slight deceleration to 10.4 percent in 2023, reflecting the incorporation of oil and gas production. Over the long run, real GDP growth is projected at 5.1 percent over the period 2020-34, slightly lower than in the last DSA. This is in line with international experience that suggests that over a long period, as economies converge to middle income status, the real growth rate slows down.
- **Fiscal deficit**. The public sector deficit is projected at 5.1 percent of GDP in 2018 and 3.7 percent over the medium term. In the long run, the deficit is set at 2.8 percent of GDP, incorporating some efforts to increase revenues, mainly through improved revenue administration and lower tax exemptions, in support of investment needs and other development challenges.
- **Current account deficit**. The current account deficit is projected to stay above 7 percent of GDP in 2018, due to an increase in imports of capital goods and oil, and to lower export growth. The current account deficit is projected to widen through 2021 due to oil and gas investment in the pre-production period. Over the long term, the average current account deficit is projected to decrease to 1.2 percent of GDP due to improved export growth, including from the oil and gas sector and the ongoing export-oriented investment in the special economic zones (SEZs). Remittances remain a significant component of the current account, but are expected to decline as a percent of GDP over the medium term.

Table 3. Senegal: Evolution of Selected Macroeconomic Indicators, 2015-18

	2015	2016	2017	2018	Med. Term[1]	Long Term[2]
Real GDP growth						
Current DSA	6.4	6.2	7.2	6.2	8.3	5.1
Previous DSA	6.5	6.7	6.8	7.0	7.0	5.4
Overall fiscal deficit (percent of GDP)						
Current DSA[3]	3.7	3.3	2.8	5.1	3.7	2.8
Previous DSA[4]	4.8	4.2	3.7	3.5	3.1	3.0
Current account deficit (percent of GDP)						
Current DSA	5.6	4.0	7.3	7.3	7.5	1.2
Previous DSA	7.0	5.6	7.8	7.2	6.8	5.3

[1] Defined as the first 5 years of the projection period. For the current DSA update, the medium term covers the years 2018-23.
[2] Defined as the last 15 years of the projection period. For the current DSA update, the long term covers the years 2024-38.
[3] Overall fiscal deficit of General Government and Public Sector.
[4] Overall fiscal deficit of Central Government.

- **Inflation**. Inflation is projected to remain contained at under 2 percent in 2018. The GDP deflator is projected at 2.2 percent in 2018 and is expected to remain around 2 percent until 2023.
- **External financing mix and terms**. The DSA assumes that the financing mix will be consistent with a prudent borrowing strategy. Even though recent borrowing has seen an increased reliance in non-concessional borrowing, the average maturity of new debt is close to 18 years, with 7-year grace period.
- **Domestic borrowing.** Domestic debt is assumed to account for 16.4 percent of total public debt on average over 2018-23, 6.2 percent of which has maturity below one year. New short-term domestic debt is assumed to be issued at an average real interest rate of 4.0 percent, while medium- and long-term domestic debt is assumed to carry a real interest rate of 4.7 percent with average maturity of 6.0 years, consistent with the current structure of domestic debt.
- **Discount rate**. The discount rate for this DSA is set at 5 percent.

Table 4. Senegal: Debt Carrying Capacity

Debt Carrying Capacity and Thresholds

Country	Senegal
Country Code	722
Debt Carrying Capacity	Strong

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Strong	Strong 3.32	Strong 3.35	Strong 3.80

Note: Until the October 2018 WEO vintage is released, the previous vintage classification and corresponding score are based solely on the CPIA per the previous framework.

Applicable thresholds

APPLICABLE	
EXTERNAL debt burden thresholds	
PV of debt in % of	
Exports	240
GDP	55
Debt service in % of	
Exports	21
Revenue	23

APPLICABLE	
TOTAL public debt benchmark	
PV of total public debt in percent of GDP	70

Calculation of the CI Index

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.818	1.47	44%
Real growth rate (in percent)	2.719	6.364	0.17	5%
Import coverage of reserves (in percent)	4.052	40.208	1.63	49%
Import coverage of reserves^2 (in percent)	-3.990	16.167	-0.65	-19%
Remittances (in percent)	2.022	10.380	0.21	6%
World economic growth (in percent)	13.520	3.579	0.48	15%
CI Score			3.32	100%
CI rating			Strong	

New framework	Cut-off values			
Weak	CI ≤	2.69		
Medium	2.69	< CI ≤	3.05	
Strong	CI >	3.05		

Reference: Thresholds by Classiciation

EXTERNAL debt burden thresholds	Weak	Medium	Strong
PV of debt in % of			
Exports	140	180	240
GDP	30	40	55
Debt service in % of			
Exports	10	15	21
Revenue	14	18	23

TOTAL public debt benchmark	Weak	Medium	Strong
PV of total public debt in percent of GDP	35	55	70

Source: Staff estimates.

EXTERNAL DSA

6. Similar to the last DSA, external debt indicators breach their thresholds for a short period of time under most stress tests (Figure 2). The external debt to export ratio breaches its threshold only under the most extreme shock. The debt service to exports ratio breaches the threshold twice, each short-lived, lasting one year: (i) in 2018 due to an operation in the first half of 2018 to reduce the Eurobond repayment spike in 2021 by 40 percent, and (ii) a more significant breach in 2021 due to the same Eurobond bullet repayment which occurs under a shock to exports.[4] Triggered tailored stress tests and the market financing risk indicators show no breach of the benchmarks or thresholds (Figure 6). A number of factors have helped counter the mechanical deterioration in indicators caused by the expanded debt coverage, including: (i) GDP rebasing, (ii) repurchase of 40 percent of the 2011 Eurobond and the resulting reduction in the 2021 bullet repayment spike, and (iii) introduction of oil and gas-related production and revenues starting in 2022 and the resulting higher exports and GDP growth.

7. Public external debt is projected to be on a downward path. Public and publicly guaranteed (PPG) external debt is estimated at 47.6 percent of GDP in 2018 and is projected to decline to 35.7 percent

[4] The start of oil and gas exports, projected in 2022 in this scenario, contributes to a significant improvement in the debt to export and debt service to export ratios.

of GDP in 2023 and below 30 percent in the long term. This baseline scenario assumes good progress on reforms, where the government takes measures to contain Treasury financing, improve revenue mobilization, and sustain growth. However, the historical scenario in the DSA provides an illustration of how lack of progress on reforms and a return to the low growth of the past would put debt on an upward and unsustainable path (Figure 2). Under this scenario, Senegal would grow at 4.5 percent and most indicators would breach their respective thresholds over the medium to long term. The historical scenario highlights the importance of steadfast implementation of structural reforms to sustain high growth.

8. Senegal remains at a low external risk of debt distress, but it is a borderline case. There are two export-related breaches for external debt and debt service indicators under the most extreme shock. Yet, mitigating factors favor maintaining the low external risk of debt distress.[5] These include:

- Authorities' ongoing exploration of debt liability management options to reduce the Eurobond bullet repayment in 2021 by 50 percent. Simulation of the debt operation under consideration would reduce the 2021 debt service spike by close to 11 percent, making the new breach less than 10 percent of the threshold, without causing new breaches over the projection period;
- Partial credit guarantee for currency risk hedging made available to Senegal by the African Development Bank and further supplementary partial guarantees for currency risk being negotiated with commercial banks;
- Unrestricted access to a pool of liquid financial assets (e.g. WAEMU international reserves currently over $15 billion or more than 4 months of regional imports);
- A sound track record of market access as evidenced by sustained and successful Eurobond issuances in 2011, 2014, 2017, and 2018, with the 2018 issuance doubling the 2017 amount and five times oversubscribed. On the regional WAEMU bond market, except for 2017 and 2018 when Senegal raised substantial financing through Eurobonds, it has also been an active market participant, with bond issuance surpassing 5 percent of GDP annually prior to 2017; and
- Authorities' ongoing structural reforms to boost private investment, especially the establishment of SEZs through the "Prosperity Triangle" project with export-oriented investment, are expected to boost exports, thus further minimizing the significance of the export-related breaches.

PUBLIC DSA

9. Indicators of overall public debt and debt service do not point to significant vulnerabilities. Total public debt increases from 60.6 percent in 2017 to 64.5 percent in 2018, and then is projected to

[5] For all public debt vulnerability indicators, the most extreme shock is the one that yields the highest ratio in or before 2028.

decline to 40.5 percent of GDP by 2038 (Figure 3 and Table 6).[6] Overall, these ratios are higher than the ones estimated in the last DSA, reflecting the broader coverage of public sector debt, increased borrowing to finance below the line Treasury operations, and greater reliance on non-concessional debt. Under the most extreme stress test, a shock to GDP growth results in a prolonged breach of the threshold, but the stress test may overstate the magnitude of the debt vulnerability in the case of Senegal because the sizable permanent increase in the primary deficit created by the standardized simulation is not consistent with Senegal's stated fiscal policy or historical data.[7] Under the historical scenario, the PV of total public debt is on an upward trend and only breaches the threshold in the last year of the projection period. Like the external DSA, the historical scenario illustrates the risk that insufficient reforms accompanied by low growth represent for public debt dynamics. Overall, risks to public debt sustainability remain low, but stress tests underline the importance of making continuous efforts to reduce the fiscal deficit, increase revenue, sustain strong economic private sector-led growth and strictly control the volumes and terms of non-concessional borrowing.

10. The public DSA highlights vulnerabilities related to debt service which are also presented in the external DSA. Debt service reaches 40.9 percent in 2018 of revenues, but falls over the medium term to 30.5 percent with some spikes due to Eurobond bullet repayments.[8] Under the current trajectory, debt service will absorb a sizable portion of fiscal revenues, thus limiting room for other expenditures in critical areas such as infrastructure investment, health, and education. As in the external DSA, this illustrates the importance of implementing key policies and reforms both to control spending and mobilize additional revenue to finance economic development in a sustainable manner.

CONCLUSION

11. According to staff's assessment, Senegal remains at low risk of debt distress, but it is a borderline case. The breach of the debt service-to-export threshold in 2021, which is due a Eurobond bullet repayment, is the most significant. Breaches of other thresholds are short-lived and do not raise significant concerns owing to the mitigating factors highlighted above, including ongoing debt liability management, guarantees to address currency risk, access to liquid financial assets, and a sound track record of market access. However, the evolution of debt ratios under the historical scenario indicates that debt sustainability hinges on continuing fiscal consolidation and on steadfastly implementing reforms to achieve high and sustained growth, as envisaged in the PSE. Senegal remains a borderline case and slippage on structural reforms or fiscal consolidation could place it at moderate risk of debt distress. Also, while progress has been made on expanding debt perimeter, the debt coverage could be improved

[6] Public debt covers central government debt and preliminary data on para-public entities and on SOEs, contrary to the last DSA which just focused on central government data.

[7] The public sector debt to GDP breach of the threshold under a shock to GDP has a large and sustained effect throughout the projection period by creating a large permanent increase in the primary deficit—about 4 percentage points of GDP higher than the baseline in 2021 and beyond. The historical scenario is well below the most extreme shock scenario, providing further evidence that the risks associated with the most extreme scenario may be overstated in the case of Senegal.

[8] As noted earlier, debt service in 2018 includes a one-off operation to reduce by 40 percent the Eurobond bullet repayment in 2021.

further. The authorities remain committed to expanding the coverage of public sector deficits and debt which will contribute to strengthening further the analysis of debt vulnerabilities.

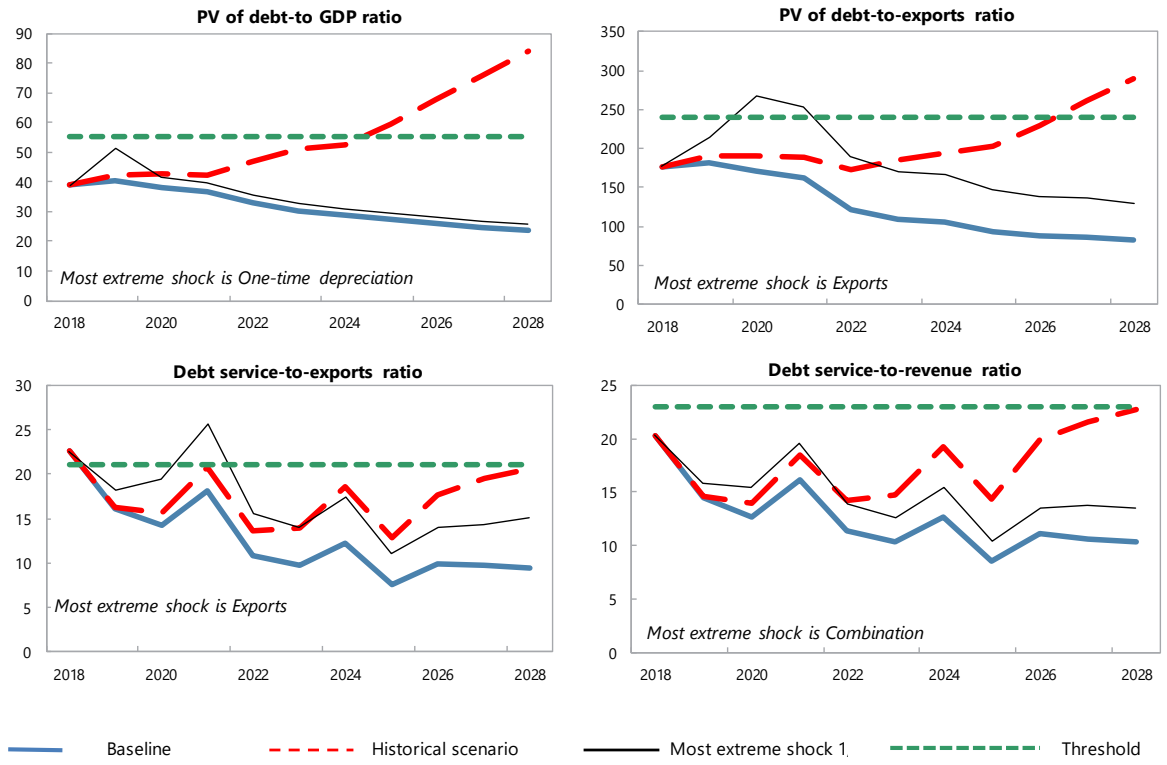
12. The current projections highlight that rising levels of debt and debt service require a cautious approach to commercial borrowing. Debt service ratios are projected to remain high despite their decline over the medium term, signaling that a further deterioration of borrowing terms could increase debt-related vulnerabilities. In this context, staff recommends a careful and continuous monitoring of financing needs and of borrowing plans, the development of a transparent pipeline of bankable projects, and a strengthening of debt management.

13. The authorities are committed to reduce debt ratios over the medium term. They aim to achieve this through further fiscal consolidation, improvement in the current account and a strengthening of debt management policies—including more attention to the terms and volume of non-concessional external financing. While progress has been made, further reforms that will effectively reduce the structural deficits of the Post Office and civil service pensions are needed, as well as addressing the current stock of appropriations from past budgets in the *comptes de dépôt*. It will be important to contain fiscal pressures from Treasury operations and address fiscal risks from the broader public sector, including the energy sector. The successful Eurobond issuance in 2018, following that of the 2017, on reasonable terms[9] indicates continued market confidence in the economy, but recent trends in bond prices suggest that future issuances could be at higher interest rates.

14. The authorities agree with the assessment in this DSA. The findings of this DSA were shared with the authorities, who broadly concurred with the staff's assessment and maintaining the low risk of debt distress rating. They plan to continue efforts to expand the coverage of public sector debt. Accordingly, they concurred with staff on the need for a prudent debt management strategy and expressed their commitment to limit non-concessional borrowing, while ensuring that borrowing decisions consider their impact on growth.

[9] The 2018 issuance was five times oversubscribed and consisted of two bonds: (i) a 30-year bond of US$1 billion, 6.75 percent interest rate, principal repayment in the last 3 years; and (ii) a 10-year bond of € 1 billion, 4.75 percent interest rate, principal repayment in the last 3 years.

Figure 2. Senegal: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2018-28



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	n.a.	n.a.
Market Financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	4.0%	4.0%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	17	17
Avg. grace period	7	7

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Senegal authorities; and staff estimates and projections.

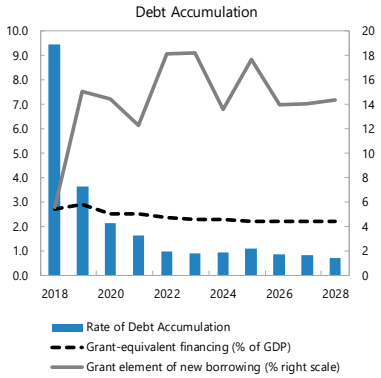
1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.
2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

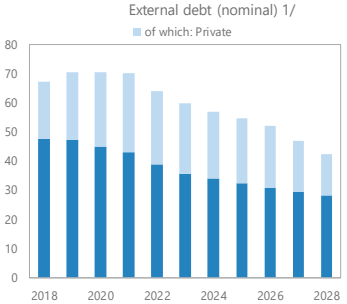
Table 5. Senegal: External Debt Sustainability Framework, Baseline Scenario, 2015-38

(Percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
External debt (nominal) 1/	**57.2**	**57.3**	**65.6**	**67.3**	**70.5**	**70.4**	**70.0**	**64.0**	**59.9**	**42.3**	**22.0**	**49.8**	**59.5**
of which: public and publicly guaranteed (PPG)	**32.3**	**33.6**	**40.4**	**47.6**	**47.2**	**44.9**	**43.0**	**38.7**	**35.7**	**28.3**	**20.0**	**27.0**	**37.4**
Change in external debt	1.7	0.1	8.3	1.7	3.3	-0.1	-0.4	-6.0	-4.1	-4.6	-1.0		
Identified net debt-creating flows	**9.8**	**-0.9**	**-1.0**	**1.8**	**-0.1**	**1.5**	**1.9**	**-6.3**	**-5.2**	**-4.8**	**-1.5**	**3.5**	**-2.5**
Non-interest current account deficit	**4.4**	**2.6**	**6.1**	**5.2**	**5.4**	**7.9**	**8.4**	**2.3**	**2.2**	**-1.5**	**0.2**	**5.8**	**2.6**
Deficit in balance of goods and services	13.0	11.0	13.6	13.8	14.3	16.8	17.3	8.3	5.7	3.0	5.0	**14.7**	**8.4**
Exports	22.4	21.5	21.7	22.0	22.3	22.4	22.6	27.1	27.7	29.0	27.8		
Imports	35.4	32.5	35.3	35.8	36.6	39.2	39.9	35.4	33.5	31.9	32.7		
Net current transfers (negative = inflow)	-9.7	-9.6	-9.1	-9.5	-9.6	-9.3	-9.2	-8.1	-7.5	-7.0	-7.0	**-9.6**	**-8.1**
of which: official	-0.5	-0.4	-0.4	-0.4	-0.3	-0.3	-0.3	-0.3	-0.3	-0.4	-0.5		
Other current account flows (negative = net inflow)	1.1	1.2	1.6	0.9	0.7	0.4	0.3	2.1	4.0	2.5	2.3	**0.8**	**2.4**
Net FDI (negative = inflow)	**-2.1**	**-1.3**	**-2.4**	**-2.0**	**-3.1**	**-4.1**	**-4.3**	**-3.9**	**-3.6**	**-2.4**	**-1.6**	**-1.7**	**-3.0**
Endogenous debt dynamics 2/	**7.5**	**-2.2**	**-4.7**	**-1.4**	**-2.3**	**-2.4**	**-2.2**	**-4.8**	**-3.7**	**-0.9**	**-0.1**		
Contribution from nominal interest rate	1.2	1.5	1.2	2.1	2.1	2.4	2.4	2.2	2.2	1.4	0.9		
Contribution from real GDP growth	-3.9	-3.3	-3.7	-3.6	-4.4	-4.8	-4.6	-7.1	-5.9	-2.2	-1.0		
Contribution from price and exchange rate changes	10.3	-0.3	-2.1	...	...	...	...	...	...	...	...		
Residual 3/	**-8.1**	**1.0**	**9.2**	**-0.1**	**3.3**	**-1.6**	**-2.3**	**0.3**	**1.1**	**0.1**	**0.5**	**-0.2**	**0.4**
of which: exceptional financing	1.7	2.3	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**34.7**	**38.8**	**40.2**	**38.2**	**36.5**	**32.7**	**30.0**	**23.9**	**17.1**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**159.7**	**176.6**	**180.7**	**170.5**	**161.5**	**120.5**	**108.1**	**82.4**	**61.6**		
PPG debt service-to-exports ratio	**7.5**	**10.4**	**8.9**	**22.6**	**16.1**	**14.3**	**18.1**	**10.9**	**9.7**	**9.4**	**6.8**		
PPG debt service-to-revenue ratio	**9.8**	**12.0**	**7.8**	**20.3**	**14.5**	**12.6**	**16.2**	**11.4**	**10.3**	**10.4**	**7.3**		
Gross external financing need (Billion of U.S. dollars)	0.9	0.9	1.4	2.3	1.7	2.3	2.9	0.9	0.9	0.3	1.3		
Key macroeconomic assumptions													
Real GDP growth (in percent)	6.4	6.2	7.2	6.2	6.9	7.5	7.1	11.6	10.4	5.1	4.5	**4.6**	**7.2**
GDP deflator in US dollar terms (change in percent)	-15.6	0.5	3.9	7.1	-1.3	3.3	1.9	2.8	1.6	1.7	2.0	**-0.2**	**2.3**
Effective interest rate (percent) 4/	2.0	2.7	2.3	3.7	3.3	3.8	3.7	3.7	3.8	3.1	4.0	**1.9**	**3.5**
Growth of exports of G&S (US dollar terms, in percent)	-7.6	2.9	12.2	15.1	6.9	11.6	10.1	37.7	14.6	7.4	7.4	**5.2**	**12.9**
Growth of imports of G&S (US dollar terms, in percent)	-13.4	-1.8	20.9	15.2	7.9	18.9	11.0	1.9	5.8	10.5	6.8	**4.6**	**8.7**
Grant element of new public sector borrowing (in percent)	...	...	...	5.6	15.1	14.4	12.3	18.1	18.2	14.3	14.9	**...**	**14.3**
Government revenues (excluding grants, in percent of GDP)	17.1	18.6	25.0	24.4	24.8	25.3	25.4	25.9	26.2	26.2	26.1	**17.2**	**25.7**
Aid flows (in Billion of US dollars) 5/	0.9	0.9	1.0	0.5	0.5	0.6	0.7	0.7	0.8	1.1	2.2		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	2.7	2.9	2.5	2.5	2.4	2.3	2.2	2.3	**...**	**2.4**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	19.7	36.6	44.6	40.4	53.3	54.4	50.9	58.8	**...**	**45.4**
Nominal GDP (Billion of US dollars)	18	19	21	24	25	28	31	35	39	58	111		
Nominal dollar GDP growth	-10.2	6.8	11.3	13.7	5.5	11.1	9.1	14.6	12.1	6.9	6.7	**4.3**	**9.7**
Memorandum items:													
PV of external debt 7/	...	...	59.8	58.5	63.6	63.6	63.5	57.9	54.1	37.9	19.0		
In percent of exports	...	...	275.4	266.1	285.6	284.3	280.9	213.5	195.2	130.8	68.6		
Total external debt service-to-exports ratio	12.2	14.6	12.7	27.9	19.7	19.5	23.3	14.9	13.6	14.8	9.1		
PV of PPG external debt (in Billion of US dollars)			7.3	9.3	10.2	10.7	11.2	11.5	11.8	13.9	18.9		
(PVt-PVt-1)/GDPt-1 (in percent)				9.5	3.7	2.1	1.7	1.0	0.9	0.7	0.6		
Non-interest current account deficit that stabilizes debt ratio	2.7	2.5	-2.2	3.5	2.1	8.0	8.8	8.3	6.3	3.1	1.3		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	Yes





Sources: Senegal authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

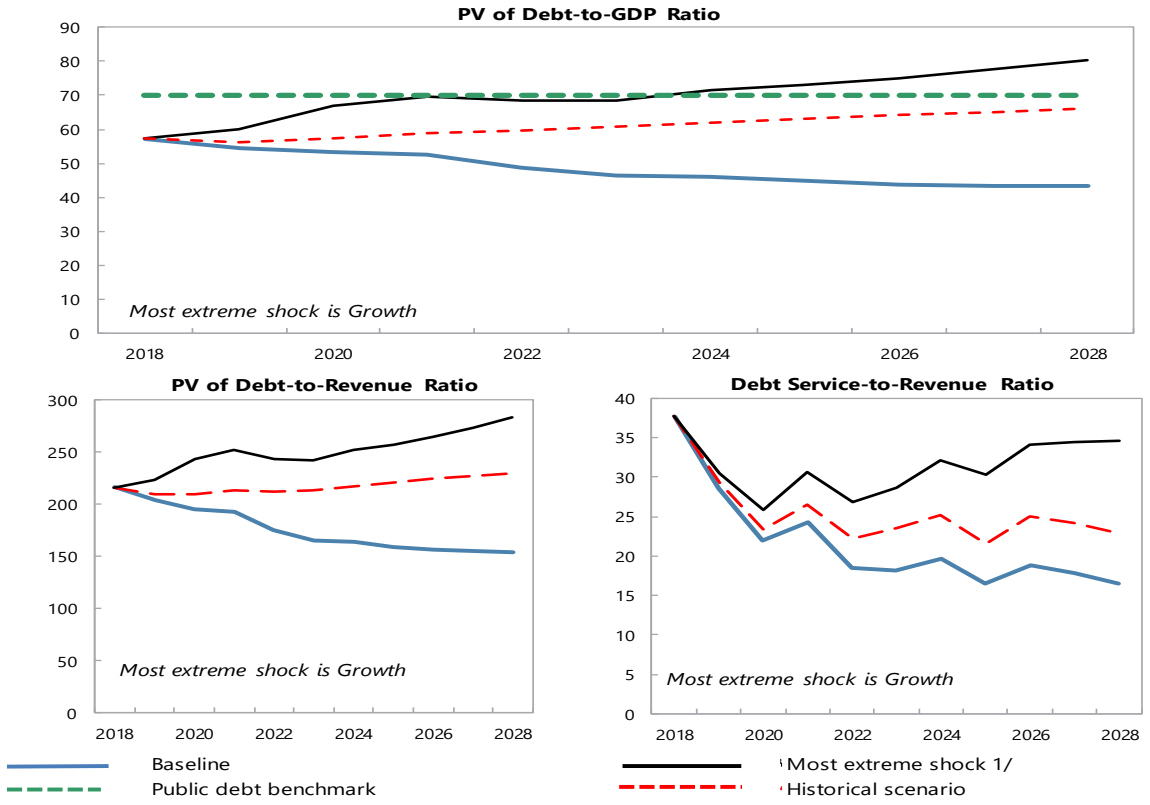
5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 3. Senegal: Indicators of Public Debt Under Alternative Scenarios, 2018-28



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	49%	49%
Domestic medium and long-term	45%	45%
Domestic short-term	5%	6%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	4.0%	4.0%
Avg. maturity (incl. grace period)	17	17
Avg. grace period	7	7
Domestic MLT debt		
Avg. real interest rate on new borrowing	1.4%	1.4%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	4%	4%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Senegal authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

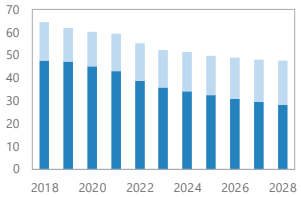
Table 6. Senegal: Public Sector Debt Sustainability Framework, Baseline Scenario, 2015-38

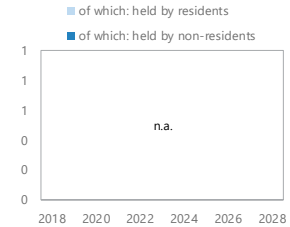
(Percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
Public sector debt 1/	**44.5**	**47.7**	**60.6**	**64.5**	**61.9**	**60.2**	**59.2**	**55.0**	**52.2**	**47.6**	**40.5**	**37.3**	**54.4**
of which: external debt	**32.3**	**33.6**	**40.4**	**47.6**	**47.2**	**44.9**	**43.0**	**38.7**	**35.7**	**28.3**	**20.0**	**27.0**	**37.4**
Change in public sector debt	2.1	3.2	13.0	3.9	-2.6	-1.7	-1.0	-4.2	-2.8	-0.5	-0.9		
Identified debt-creating flows	**3.9**	**-0.6**	**-6.5**	**1.8**	**-1.0**	**-1.7**	**-0.9**	**-4.2**	**-2.8**	**-0.5**	**-0.9**	**2.0**	**-1.2**
Primary deficit	**2.2**	**1.6**	**0.9**	**3.1**	**2.3**	**1.7**	**1.7**	**1.0**	**1.1**	**1.2**	**0.7**	**2.6**	**1.6**
Revenue and grants	19.3	20.7	27.1	26.5	26.8	27.2	27.4	27.8	28.1	28.1	28.1	**19.3**	**27.7**
of which: grants	2.2	2.2	2.2	2.1	2.0	2.0	2.0	1.9	1.9	1.9	2.0		
Primary (noninterest) expenditure	21.4	22.4	28.0	29.6	29.2	28.9	29.0	28.9	29.1	29.2	28.8	**21.9**	**29.2**
Automatic debt dynamics	**2.2**	**-0.3**	**-6.0**	**-0.6**	**-3.3**	**-3.4**	**-2.6**	**-5.3**	**-3.8**	**-1.7**	**-1.6**		
Contribution from interest rate/growth differential	-1.8	-1.4	-2.5	-2.1	-2.8	-2.9	-2.6	-5.0	-3.9	-1.2	-1.2		
of which: contribution from average real interest rate	0.8	1.2	0.7	1.4	1.3	1.4	1.4	1.2	1.2	1.2	0.6		
of which: contribution from real GDP growth	-2.5	-2.6	-3.2	-3.5	-4.2	-4.3	-4.0	-6.1	-5.2	-2.3	-1.8		
Contribution from real exchange rate depreciation	4.0	1.1	-3.4	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**-0.5**	**-1.9**	**-1.4**	**-0.7**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**-0.3**	**-0.1**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	-0.5	-1.9	-1.4	-0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-1.7**	**3.8**	**19.5**	**3.6**	**-2.1**	**-0.4**	**-0.1**	**-0.3**	**0.1**	**-0.5**	**-0.4**	**2.3**	**-0.2**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**53.3**	**57.2**	**54.7**	**53.3**	**52.6**	**48.9**	**46.4**	**43.2**	**37.6**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**196.5**	**216.3**	**203.9**	**195.7**	**192.3**	**175.7**	**165.3**	**153.8**	**133.8**		
Debt service-to-revenue and grants ratio 3/	**30.1**	**35.6**	**25.9**	**37.7**	**28.6**	**22.0**	**24.4**	**18.4**	**18.2**	**16.5**	**11.9**		
Gross financing need 4/	6.8	7.4	8.0	12.3	10.0	7.7	8.4	6.1	6.2	5.8	4.0		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	6.4	6.2	7.2	6.2	6.9	7.5	7.1	11.6	10.4	5.1	4.5	**4.6**	**7.2**
Average nominal interest rate on external debt (in percent)	2.3	3.2	2.5	4.5	3.8	4.1	4.1	4.1	4.3	3.6	3.5	**2.2**	**3.9**
Average real interest rate on domestic debt (in percent)	3.9	5.3	3.6	3.1	3.8	3.2	3.4	2.1	2.7	0.9	-0.4	**3.5**	**2.1**
Real exchange rate depreciation (in percent, + indicates depreciation)	13.4	3.5	-10.8	...	...	...	...	...	...	...	...	**2.6**	**...**
Inflation rate (GDP deflator, in percent)	1.1	0.8	1.8	2.3	1.8	2.0	1.3	2.0	1.1	1.7	2.0	**1.4**	**1.8**
Growth of real primary spending (deflated by GDP deflator, in percent)	4.8	10.8	34.2	12.1	5.5	6.7	7.5	10.8	11.4	4.8	4.4	**7.9**	**7.7**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	0.0	-1.5	-12.1	-0.7	5.0	3.4	2.6	5.3	3.8	1.7	1.6	**-4.5**	**2.7**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**Yes**

Public sector debt 1/





Sources: Senegal authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments plus extra budgetary funds, central bank, government-guaranteed debt, non-guaranteed SOE debt. Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 7. Senegal: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2018-28

(Percent)

	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of debt-to GDP ratio											
Baseline	39	40	38	36	33	30	29	27	26	25	24
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	39	42	43	42	47	51	53	**59**	**68**	**76**	**84**
B. Bound Tests											
B1. Real GDP growth	39	43	44	42	38	35	33	31	30	29	28
B2. Primary balance	39	41	40	38	35	32	31	29	28	27	26
B3. Exports	39	43	46	44	40	36	35	33	31	30	29
B4. Other flows 3/	39	42	43	41	37	34	32	31	29	28	27
B5. Depreciation	39	51	42	40	36	33	31	29	28	27	26
B6. Combination of B1-B5	39	48	49	47	42	39	37	35	33	32	31
C. Tailored Tests											
C1. Combined contingent liabilities	39	43	41	40	37	34	33	31	30	29	28
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	39	45	43	41	37	34	32	30	29	27	26
Threshold	55	55	55	55	55	55	55	55	55	55	55
PV of debt-to-exports ratio											
Baseline	177	181	170	161	120	108	106	92	87	86	82
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	177	190	191	188	172	184	194	202	229	**262**	**289**
B. Bound Tests											
B1. Real GDP growth	177	181	170	161	120	108	106	92	87	86	82
B2. Primary balance	177	183	178	170	127	115	113	99	94	92	89
B3. Exports	177	215	**267**	**253**	190	170	167	146	139	136	129
B4. Other flows 3/	177	189	191	181	135	122	119	104	99	97	92
B5. Depreciation	177	181	147	139	103	92	90	79	74	73	71
B6. Combination of B1-B5	177	211	187	212	158	142	139	122	116	113	108
C. Tailored Tests											
C1. Combined contingent liabilities	177	195	184	177	135	122	120	105	100	99	95
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	177	181	171	162	121	108	106	92	87	85	81
Threshold	240	240	240	240	240	240	240	240	240	240	240
Debt service-to-exports ratio											
Baseline	**23**	16	14	18	11	10	12	8	10	10	9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	**23**	16	16	21	14	14	19	13	18	20	21
B. Bound Tests											
B1. Real GDP growth	**23**	16	14	18	11	10	12	8	10	10	9
B2. Primary balance	**23**	16	14	19	11	10	13	8	10	10	10
B3. Exports	**23**	18	19	**26**	16	14	17	11	14	14	15
B4. Other flows 3/	**23**	16	15	19	11	10	13	8	10	11	11
B5. Depreciation	**23**	16	14	17	10	9	12	7	9	9	8
B6. Combination of B1-B5	**23**	17	18	**23**	14	12	15	10	12	13	12
C. Tailored Tests											
C1. Combined contingent liabilities	**23**	16	15	19	11	10	13	8	10	10	10
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	**23**	16	15	20	13	11	14	10	11	9	9
Threshold	21	21	21	21	21	21	21	21	21	21	21
Debt service-to-revenue ratio											
Baseline	20	14	13	16	11	10	13	8	11	11	10
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	20	15	14	18	14	15	19	14	20	22	23
B. Bound Tests											
B1. Real GDP growth	20	16	15	19	13	12	15	10	13	12	12
B2. Primary balance	20	14	13	17	12	11	13	9	11	11	11
B3. Exports	20	15	13	18	13	11	14	9	12	12	13
B4. Other flows 3/	20	14	13	17	12	11	13	9	12	12	12
B5. Depreciation	20	18	16	19	13	12	15	10	13	13	11
B6. Combination of B1-B5	20	16	15	20	14	13	15	10	13	14	13
C. Tailored Tests											
C1. Combined contingent liabilities	20	14	13	17	12	11	13	9	12	11	11
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	20	14	13	17	13	12	15	11	12	10	9
Threshold	23	23	23	23	23	23	23	23	23	23	23

Sources: Senegal authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 8. Senegal: Sensitivity Analysis for Key Indicators of Public Debt, 2018-28

(Percent of GDP, unless otherwise indicated)

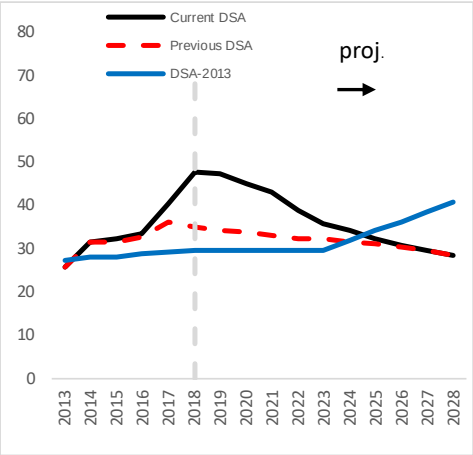
	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of Debt-to-GDP Ratio											
Baseline	57	55	53	53	49	46	46	45	44	43	43
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	57	56	58	59	60	61	62	63	64	65	66
B. Bound Tests											
B1. Real GDP growth	57	60	67	70	69	69	71	73	75	78	80
B2. Primary balance	57	56	56	56	52	49	48	47	46	46	45
B3. Exports	57	57	60	59	55	52	52	50	49	48	48
B4. Other flows 3/	57	57	58	57	53	50	50	48	47	47	46
B5. Depreciation	57	64	60	57	51	46	44	41	38	36	34
B6. Combination of B1-B5	57	55	57	57	53	51	52	51	51	51	51
C. Tailored Tests											
C1. Combined contingent liabilities	57	61	60	59	54	51	51	49	48	48	47
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	57	55	53	53	49	46	46	45	44	43	43
Public debt benchmark	70	70	70	70	70	70	70	70	70	70	70
PV of Debt-to-Revenue Ratio											
Baseline	216	204	196	192	176	165	164	159	156	155	154
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	216	209	210	213	212	213	218	221	224	227	230
B. Bound Tests											
B1. Real GDP growth	216	223	244	252	244	242	252	257	264	274	283
B2. Primary balance	216	208	207	203	185	174	172	168	164	163	161
B3. Exports	216	212	221	217	198	186	183	178	175	172	169
B4. Other flows 3/	216	211	212	208	190	179	176	172	168	166	164
B5. Depreciation	216	241	222	210	184	166	157	146	137	129	121
B6. Combination of B1-B5	216	204	208	207	192	183	183	181	180	180	181
C. Tailored Tests											
C1. Combined contingent liabilities	216	229	219	214	195	183	181	176	172	170	169
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	216	204	196	193	176	166	163	159	155	154	153
Debt Service-to-Revenue Ratio											
Baseline	38	29	22	24	18	18	20	17	19	18	16
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	38	29	23	27	22	23	25	22	25	24	23
B. Bound Tests											
B1. Real GDP growth	38	30	26	31	27	29	32	30	34	34	35
B2. Primary balance	38	29	22	26	21	20	21	18	20	19	18
B3. Exports	38	29	22	25	19	19	20	17	19	19	19
B4. Other flows 3/	38	29	22	25	19	19	20	17	19	19	18
B5. Depreciation	38	29	25	28	21	20	22	18	21	19	17
B6. Combination of B1-B5	38	28	23	26	21	21	23	20	23	22	21
C. Tailored Tests											
C1. Combined contingent liabilities	38	29	24	30	24	21	22	19	20	19	18
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	38	29	23	25	20	20	21	19	20	17	16

Sources: Senegal authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

Figure 4. Senegal: Driver of Debt Dynamics—Baseline Scenario, 2013-28

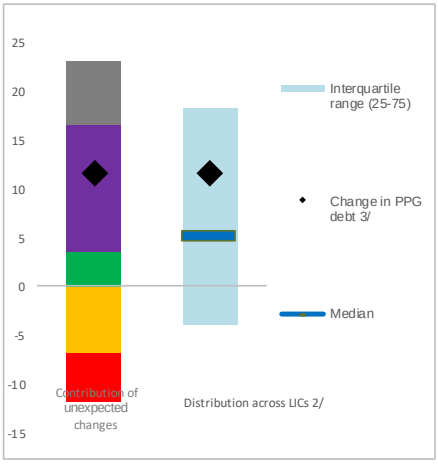
External debt

Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



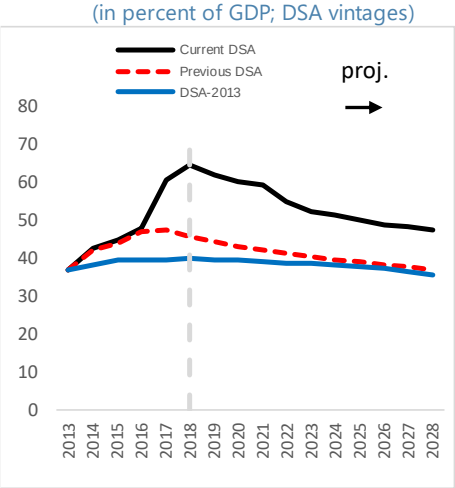
Debt-creating flows
(percent of GDP)

Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

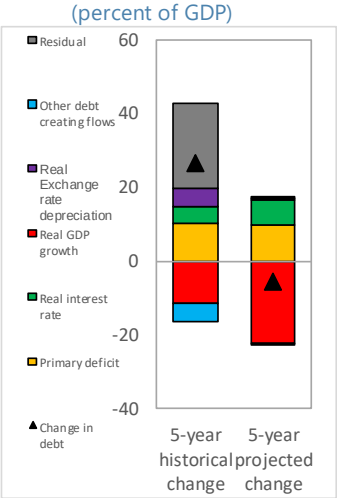


Public debt

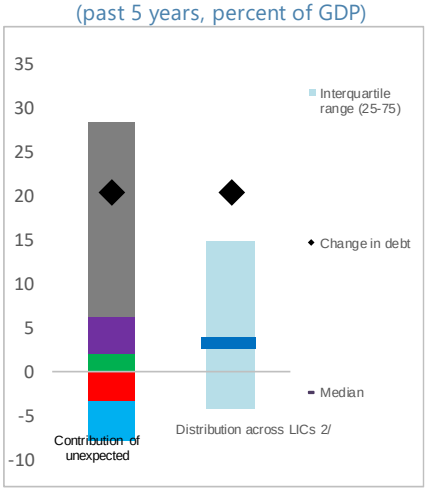
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



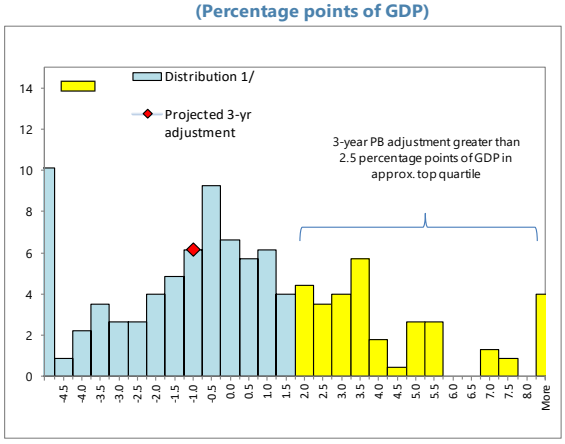
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

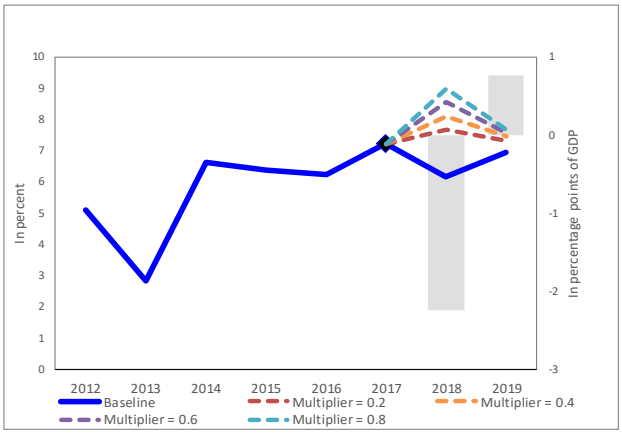
Figure 5. Senegal: Realism Tools, 2012-2023

3-Year Adjustment in Primary Balance (Percentage points of GDP)



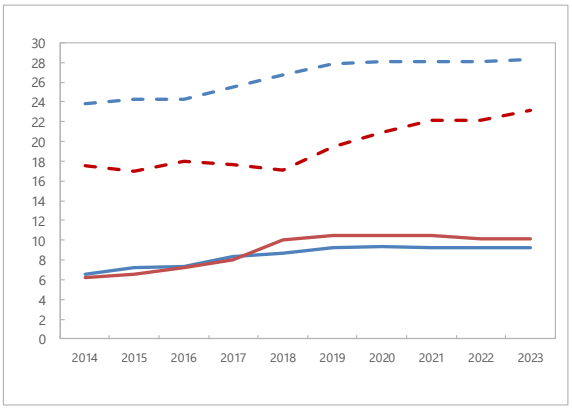
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/

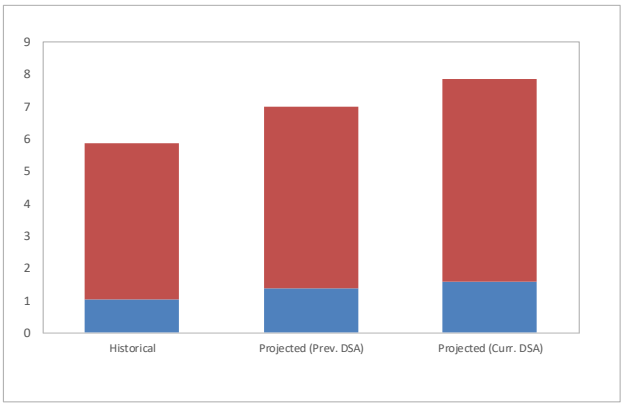


1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



Contribution to Real GDP growth
(percent, 5-year average)



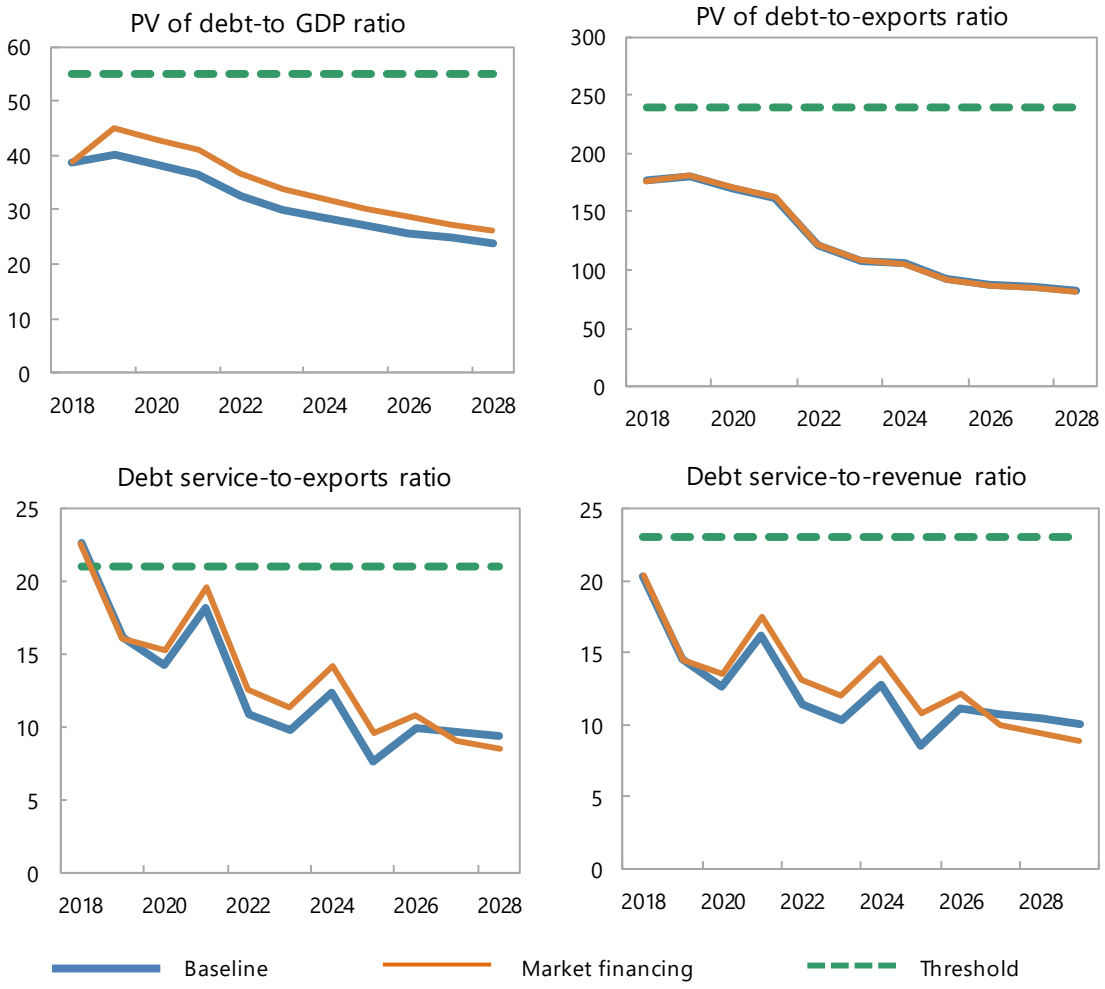
Gov. Invest. - Prev. DSA
Gov. Invest. - Current DSA
Priv. Invest. - Prev. DSA
Priv. Invest. - Current DSA
Contribution of other factors
Contribution of government capital

Source: Staff estimates.

Figure 6. Senegal: Market-Financing Risk Indicators, 2018-28

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	12	470
Breach of benchmark	No	No
Potential heightened liquidity needs	Low	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



Sources: Senegal authorities; and staff estimates and projections.